

README document for manuscript ” A flexible Bayesian framework for detecting cross-sample spatial expression variability in heterogeneous tissues”

Data

All datasets (real spatial transcriptomic data and Simulation data) used in this study are publicly available and stored in the GitHub repository [zhoudeng123456/IBaySVG-analysis](https://github.com/zhoudeng123456/IBaySVG-analysis).

Directory Structure Description

The data used in this study is stored in the `data/` folder of the repository, which is divided into two core subdirectories:

- **Realdataset:** Contains the two publicly available spatial transcriptomic datasets described below, including both raw and preprocessed gene expression matrices, benchmark scRNA/snRNA-seq reference matrices, spatial coordinates, and cell-type deconvolution proportions.
- **Simulation data:** Includes simulation datasets spanning three canonical spatial patterns and Three additional heterogeneous spatial structures. For each canonical pattern, multiple data-integration scenarios are constructed, with and without covariates, to systematically evaluate model performance under varying structural complexities.

Abstract

The spatial transcriptomic datasets used in this study include two publicly available datasets:

- Human dorsolateral prefrontal cortex (DLPFC), comprising laminar cortical structures and evaluated under within-donor and cross-donor section integration scenarios.
- Human squamous cell carcinoma (SCC), a representative tumor tissue dataset characterized by spatial heterogeneity.

These datasets differ in spot resolution (~1,000–4,000 spots), tissue architecture, and sparsity levels, providing complementary benchmarks for evaluating model performance under diverse spatial transcriptomic settings.

Availability

All datasets are publicly available for download via the following links:

- **DLPFC:** The raw datasets used in this study include both spatial transcriptomics (ST) and single-nucleus RNA-seq (snRNA-seq) references (Maynard et al., 2021). URLs:
 - ST Dataset: Available at the GitHub repository <https://github.com/LieberInstitute/HumanPilot>
 - snRNA-seq Reference: https://www.dropbox.com/s/5919zt00vm1ht8e/sce_DLPFC_annotated.zip?dl=1.
- **SCC:** Available at the Gene Expression Omnibus (GEO) under accession number GSE144240 and GSE144236 (Ji et al., 2020). URLs:
 - ST Dataset: <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE144240>
 - scRNA-seq Reference: <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE144236>

Real Dataset Description

Human dorsolateral prefrontal cortex (DLPFC)

- **Contributors:** Maynard, K.R. and Collado-Torres, L. , et al.
- **Citation:** Maynard, K., Collado-Torres, L. et al. (2021), ‘Transcriptome-scale spatial gene expression in the human dorsolateral prefrontal cortex’, Nature Neuroscience 24, 425–436.
- **File format:** Gene expression matrix (HDF5), spatial coordinates (TXT), snRNA-seq count matrix (HDF5), and snRNA-seq cell-type and gene annotations (RDS).
- **Preprocessing:**
 - Quality control: Cells with <500 expressed genes and genes with <100 expressing cells were filtered out.
 - Selecting the top 8,000 highly variable genes (HVGs) per slice using the Seurat package and identifying the consensus HVGs across all four slices.
 - Focused on seven cell types (Astro, EndoMural, MicroOligo, Oligo, OPC, Excit, and Inhib), with spot-wise proportions estimated using Redeconve (Zhou et al. 2023).

Human squamous cell carcinoma (SCC)

- **Contributors:** Ji, A. L., Rubin, A. J., Thrane, K., Jiang, S., Reynolds, D. L., Meyers, R. M., Guo, M., George, J., Mollbrink, A. et al.
- **Citation:** Ji, A. L., Rubin, A. J., Thrane, K., Jiang, S., Reynolds, D. L., Meyers, R. M., Guo, M., George, J., Mollbrink, A. et al. (2020), ‘Multimodal analysis of composition and spatial architecture in human squamous cell carcinoma’, Cell 182(2), 497–514.e22.
- **File format:** Gene expression matrix (TSV), spatial coordinates (TSV), scRNA-seq reference counts (TXT), and scRNA-seq reference cell-type annotations (TXT).
- **Preprocessing:**
 - Quality control: Cells with <100 expressed genes and genes with <10 expressing cells were filtered out.
 - Selecting the top 8,000 highly variable genes (HVGs) per slice using the Seurat package and identifying the consensus HVGs across all four slices.
 - Focused on three major cell types (Keratinocyte, TSK, and Melanocyte), with spot-wise proportions estimated using Redeconve.

Simulation Data Description

To further validate the performance of IBaySVG under controlled conditions, we employed simulation datasets spanning three canonical spatial patterns and nine additional heterogeneous spatial structures, which are stored in the Simulation data subdirectory. For each canonical pattern, multiple data-integration scenarios were constructed, with and without covariates, enabling systematic evaluation of the method under varying spatial architectures and integration complexities. These simulations provide a structured and challenging benchmark for assessing the accuracy and robustness of IBaySVG in identifying spatially variable genes.

Dataset	Core Characteristics
Three canonical spatial patterns	Each pattern includes 12 integration scenarios with varying signal strengths (high to low) and zero-inflation rates (high, medium, low), plus additional scenarios without covariates.

Dataset	Core Characteristics
Three additional heterogeneous spatial structures	Comprised of three classes of complex spatial architectures: (i) zero-inflated nearest-neighbor Gaussian process (ZINNGP) capturing covariance-based spatial dependencies, (ii) ZINB-NonSpa hybrid models with linear-focal, linear-periodic, and focal-periodic fusion patterns, and (iii) enhanced ZINB-NonSpa patterns including sigmoidal activation and four polynomial functional forms.

All the simulated data are generated by ZINB distributions or Gaussian process framework with covariates from Dirichlet sampling to evaluate model performance under diverse and non-canonical spatial configurations.

Reproduce the results

Abstract

The `code_and_data` directory contains all core resources to reproduce results:

- **data/**: contains all datasets used in this study, including real datasets (both processed datasets and raw input files for DLPFC and SCC studies) and simulated datasets generated under different spatial structure scenarios for benchmarking and evaluation.
- **Demo/**: the main IBaySVG implementation, usage instructions for both IBaySVG and comparison methods, and a small example dataset for testing the full analysis workflow.
- **Simulation Generate/**: Standalone code to generate synthetic spatial expression data across multiple canonical (linear, focal, periodic) and heterogeneous spatial patterns, with tunable signal strengths, zero-inflation rates, and covariate configurations.
- **Simulations/**: Scripts and results for simulation studies(including parameter sensitivity analyses).
- **RealData/**: Scripts and results for real spatial transcriptomic datasets (including stability assessments, model checking and parameter sensitivity analysis).

Instructions for Use

All results in the manuscript (simulations and real data analyses) are fully reproducible. The `code_and_data` directory is structured hierarchically to facilitate step-by-step reproduction, with strict path management via the `here` package (**critical: set the working directory root to `code_and_data` before running any scripts**).

data This directory contains real and simulated datasets for reproducibility and evaluation. See `data/readme_data.pdf` for details.

Subdirectory	Content Description
<code>Realdataset/dlpfc_samedonor/</code>	Contains processed DLPFC datasets from the same donor, including ST count matrices, spatial coordinates, and cell proportion matrices.

Subdirectory	Content Description
Realdataset/dlpfc_acrossdonor/	Contains processed DLPFC datasets across different donors, including ST count matrices, spatial coordinates, and cell proportion matrices.
Realdataset/scc/	Contains processed SCC datasets, including ST count matrices, spatial coordinates, and cell proportion matrices.
Realdataset/raw_data/	Contains the raw input files for the DLPFC and SCC datasets (the folder is left empty in the supplementary archive and corresponding files should be downloaded via GitHub).
Simulation data/basic simulation	Contains simulated datasets corresponding to the three main spatial structure settings (the folder is left empty in the supplementary archive and corresponding files should be downloaded via GitHub).
Simulation data/additional simulation	Contains simulated datasets corresponding to the nine additional spatial structure settings (the folder is left empty in the supplementary archive and corresponding files should be downloaded via GitHub).
Simulation data/basic simulation without covariates	Contains simulated datasets under the three main spatial structure settings without covariates (the folder is left empty in the supplementary archive and corresponding files should be downloaded via GitHub).

Demo The Demo/ folder provides a minimal working example for reproducing the main analysis pipeline and benchmarking procedures.

It contains the following components:

- **Implement_of_IBaySVG.pdf** Provides detailed documentation of the IBaySVG algorithm and instructions for its implementation.
- **IBaySVG_main.R** The main function implementing the proposed IBaySVG method.
- **Comparison_methods_R.R**: Implementation of competing methods in the R environment.
- **Comparison_methods_python.py**: Implementation of competing methods in the Python environment.
- **Example data files**: The remaining files provide example datasets in both R and CSV formats:
 - `data_example.RData` — R-format example dataset.
 - `matrix*_count_example.csv` — UMI count matrices (CSV format).
 - `matrix*_position_example.csv` — Spatial coordinate matrices (CSV format).

The R-formatted files are intended for use in the R environment, while the CSV files are provided to ensure compatibility with Python-based analysis.

Simulation Generate This directory contains standalone code to generate synthetic spatial expression data across multiple canonical (linear, focal, periodic) and heterogeneous spatial patterns, with tunable signal strengths, zero-inflation rates, and covariate configurations:

Exact File List	Content Description
generate_simu.R	Generates synthetic spatial transcriptomic datasets under a zero-inflated negative binomial (ZINB) model with configurable spatial effect structures: - Outputs: Spot $\times$ gene count matrix ($N \times G$) Spot $\times$ 2 spatial coordinate matrix ($N \times 2$) Optional spot $\times$ covariate matrix ($N \times J$) - Supports canonical and heterogeneous spatial patterns (e.g., linear, focal, periodic, and extended spatial structures) - Tunable parameters: Proportion of spatially variable (SV) genes Spatial signal strength for SV/non-SV genes Zero-inflation rate (dropout level) Dispersion parameter of the ZINB distribution Baseline mean expression level Covariate inclusion and distribution scenarios

To generate simulation data:

1. Set working directory to `code_and_data/Simulation Generate/`;
2. choose the spatial pattern and modify parameter values (zero-inflation, spatial signal strength, Covariate inclusion) according to the function documentation;

Simulations Updated with exact file mappings for simulation studies:

Subdirectory	Exact File List	Content Description
code/	simulation_spotregion_draw_Figure_S10.R	Generates plots for partitioned spot regions considered in the basic simulations.(Supplementary Figure S10)
	simulation_draw_Figures_S11-S19.R	Generates plots for basic simulation results under three canonical spatial patterns(Supplementary Figures S11-S19)
	additional_expression_pattern_Figure_S20.R	Generates plots for examples of spatial expressions under additional spatial structures(Supplementary Figure S20)
	simulation_additional_draw_Figures_S21-S23.R	Generates plots for simulation results under nine additional spatial structures (Supplementary Figures S21-S23)
	simulation_nocov_draw_Figures_S24-S26.R	Generates plots for simulation results under three canonical spatial patterns without covariates (Supplementary Figures S24-S26)

Subdirectory	Exact File List	Content Description
	<code>sensitive_simulation_draw_Figures_S27-S29.R</code>	Generates plots for sensitivity analysis of hyperparameters in the simulated dataset under three canonical spatial patterns(Supplementary Figures S27–S29)
	<code>comparison_between_ZINB_NB_Tables_S5-S6.R</code>	Illustrates the comparison between ZINB and NB Models under simulation scenarios for Tables S5-S6
	<code>comparison_between_two_statistics_Tables_S1-S3.R</code>	Illustrates the comparison of the performance between two statistics under simulation scenarios for Tables S1-S3
	<code>Computer_time_S4.R</code>	Illustrates the average execution time per gene under different spots numbers for Table S4
<code>result_data/</code>	<code>sensitivity analysis of hyperparameters/ (directory)</code>	Contains results from sensitivity analyses of key hyperparameters evaluated on simulated datasets under three canonical spatial patterns (linear, focal, and periodic)
	<code>simulation result/ (directory)</code>	Contains results from baseline simulation scenarios (linear, focal, and periodic patterns), evaluated both with and without covariate effects, as well as additional simulation settings under nine alternative spatial structures.
	<code>additional spatial pattern/ (directory)</code>	Contains plots of spatial expression of additional spatial structures selected (the folder is left empty in the supplementary archive and corresponding files should be downloaded via GitHub)
	<code>noinflation/(directory)</code>	Contains the code and results of IBaySVG without zero inflation structure (the example file is left empty in the supplementary archive and corresponding files should be downloaded via GitHub)
	<code>compare statistics/ (directory)</code>	Contains results of using different statistics in IBaySVG
	<code>compute time.csv</code>	Contains the results of computing time of IBaySVG

RealData This directory contains full-scale analysis results and plotting scripts for real spatial transcriptomic datasets, with the exact file breakdown as follows:

Subdirectory	Exact File List	Content Description
code/	venn_plot_draw_Figures_S1-S3.R	Generates venn plots of SV genes identified in real datasets for Supplementary Figures S1-S3
	celltype_plot_draw_Figures_S30-S32.R	Generates plots of distribution of cellular composition in real datasets for Supplementary Figures S30-S32
	Model_diagnostics_draw_Figures_S34-S39_Tables_S28-S31.R	Performs model-structure evaluations in the real datasets, including likelihood ratio tests comparing the ZINB and NB models, Vuong tests comparing the linear, focal, and periodic models, and diagnostic analyses of the B-spline structure, and generates Supplementary Tables S9-S11 and Figures S34-S39
	upset_draw_Figure_S43-S45.R	Generates upset plots of SV gene identification through multiple integration strategies in real datasets for Supplementary Figures S43-S45
	gene_cluster_draw_Figures_S48-S53_Tables_S24-S26.R	Generates expression and heatmap plots of dominant gene clusters of SV genes identified by IBaySVG in real datasets for Supplementary Figures S48-S53 and the enrichment result for supplementary Tables S24-S26
	sensitive_realdata_draw_Figures_S40-S42.R	Generates plots for sensitivity analysis of hyperparameters in real datasets (Supplementary Figures S40-S42)
	spatial_domain_draw_Figure3.R	Generates spatial domains derived from SV genes identified by IBaySVG for main Figure3
	Stability_analysis_Tables_S12-S17.R	Illustrates the results of subsampling validation of SV gene detection in real dataset for Tables S12-S17
	marker_gene_identify_Tables_S18-S20.R	Illustrates the performance comparison of different methods in identifying layer-specific marker genes and presents the corresponding marker gene lists for Tables S18-S20

Subdirectory	Exact File List	Content Description
	<code>comparison_spotcluster_index_Tables_S27-S44.R</code>	Illustrates the comparative evaluation of spot-clustering performance in the real datasets using both the complete gene sets and the reduced gene sets obtained after removing genes uniquely identified by IBaySVG for Tables S27-S44
	<code>comparative_analysis_Figures_2_S46_S47.R</code>	Illustrates the process of comparative analysis of the identified SV genes and generates the plots of spatial expression for Figures 2, S46 and S47
	<code>Exploratory_analysis_Figures_S4-S9.R</code>	Illustrates the exploratory analysis of the zero-inflation, B-spline, and bi-level structures in the proposed model for Figures S4-S9
	<code>data_preprocess.R</code>	Illustrates the comprehensive data cleaning, quality control, and preprocessing pipeline starting from the raw datasets
	<code>unique_gene_bioevidence_Tables_S21-23.R</code>	Presents the complete lists of genes uniquely identified by IBaySVG for Tables S21-S23.
	<code>outlier_detection_Tables_S7-S8_Figure_S33.R</code>	Illustrates the outlier detection analysis in the real datasets and generates Tables S7-S8 and Figure S33.
<code>result_data/</code>	<code>realdata svgene/(directory)</code>	Contains identified SV genes of each method
	<code>genecluster/(directory)</code>	Contains the results of inferred gene cluster (the folder is left empty in the supplementary archive and corresponding files should be downloaded via GitHub)
	<code>spotcluster/(directory)</code>	Contains the results of inferred spot cluster (the folder is left empty in the supplementary archive and corresponding files should be downloaded via GitHub)
	<code>stability/(directory)</code>	Contains the results of stability analysis of each method
	<code>model check/(directory)</code>	Contains the results of model checking
	<code>sensitive analysis/(directory)</code>	Contains the results of sensitive analysis of hyperparameters
	<code>upset plot/(directory)</code>	Contains the preprocessed upset plot

Subdirectory	Exact File List	Content Description
	<code>benchmark/(directory)</code>	Contains the benchmark results obtained from Zeng et al.(2012) and Maynard et al.(2021). (the folder is left empty in the supplementary archive and corresponding files should be downloaded via GitHub)
	<code>outlier_detection/(directory)</code>	Contains intermediate results generated during the outlier detection analyses
	<code>exploratory_analysis/(directory)</code>	Contains intermediate results generated during the exploratory analyses

Downloading Large Reproducibility Files Some intermediate outputs and large data files (e.g., simulation and realdata analysis outputs, and processed count matrices) are excluded from this repository due to their size. To fully reproduce the results in the paper:

1. Download the required files from: <https://github.com/zhoumeng123456/IBaySVG-analysis>
2. Specifically, download and replace the following directories:
 - `data/`
 - `RealData/result_data/benchmark/`
 - `RealData/result_data/geneccluster/`
 - `RealData/result_data/spotcluster/`
 - `Simulations/result_data/additional_spatial_pattern/`
 - `Simulations/result_data/noinflation/example_linear_inf5.RData`

Note: The GitHub repository version corresponding to the supplementary materials and ACC submission is: commit ID: 4353b68b3a00052d79ccbc88cbb7ef6d75f0b691. This commit represents the exact version used for the revised reproducibility package and supplementary materials.

3. Place the downloaded folders in the same relative directory structure as provided in this repository.
4. Do not modify folder names, as the scripts rely on relative paths.

All analysis scripts will run without modification once the directory structure is preserved.

Reproducibility Workflow

The recommended order for reproducing the results in this manuscript is:

Step 1: Download Required Files from GitHub before running any scripts

Step 2: Set the Working Directory All R scripts rely on the `here` package for path management. Set the working directory to the `code_and_data` root directory before running any scripts.

Step 3: Run the Demo Example (Quick Start) Users are encouraged to begin with the example provided in the `Demo/` directory. This minimal working example demonstrates:

- Implementation of the IBaySVG method by reading the `Implement_of_IBaySVG.pdf` and running `IBaySVG_main.R`
- Implementation of the Comparative method by running:
 - `Comparison_methods_R.R`
 - `Comparison_methods_python.py`
- Expected input and output formats.

Step 4: Reproduce Simulation Studies Simulation experiments can be reproduced using the `Simulation Generate/` and `Simulations/` directory. Users may either regenerate the simulation datasets or directly reproduce results using the provided data files.

- Custom simulation data can be generated using `Simulation Generate/generate_simu.R`;
- or load the specific simulation data from `data/Simulation data/`;
- All simulation figures and tables can be regenerated using scripts in:
 - `Simulations/code/`
 - `Simulations/result_data/`

Step 5: Reproduce Real Data Analyses All real-data analyses are contained in the `RealData/` directory. To fully reproduce the results, we recommend the following execution order:

(1) Exploratory analysis

First, proceed with exploratory analyses to examine the roles and empirical behavior of the zero-inflation, B-spline, and bi-level structures in the proposed model.

Scripts are located in:

- `RealData/code/Exploratory_analysis_Figures_S4-S9.R`
- `RealData/result_data/exploratory_analysis/`

(2) Outlier detection, Model Diagnostics, Stability analysis and Sensitive analysis

Begin with a comprehensive assessment of model reliability, including outlier detection, model checking and diagnostic analyses of distributional assumptions and spatial functional forms, subsampling-based stability assessment, and sensitivity analyses of hyperparameter settings.

Scripts are located in:

- Outlier detection:
 - `RealData/code/outlier_detection_Tables_S7-S8_Figure_S33.R`
 - `RealData/result_data/outlier detection/`
- Model Diagnostics:
 - `RealData/code/Model_diagnostics_draw_Figures_S34-S39_Tables_S9-S11.R`
 - `RealData/result_data/model check/`
- Subsampling-based stability assessment:
 - `RealData/code/Stability_analysis_Tables_S12-S17.R`
- Sensitive Analysis of hyperparameters:

- `RealData/code/sensitive_realdata_draw_Figures_S40-S42.R`

(3) Identification of Spatially Variable (SV) Genes

Running the IBaySVG and competing methods produced in `Demo/` to identify SV genes based on the pre-processed datasets provided in:

- Option A (Quick Start): Users can directly utilize the preprocessed benchmarks provided in `data/Realdataset/` for an immediate run.
- Option B (Full Reproduction): Alternatively, reproduction users can completely regenerate these preprocessed datasets from the raw matrices by running the provided data cleaning pipeline `RealData/code/data_preprocess.R`.

The resulting SV gene lists are stored in:

- `RealData/result_data/realdata_svgene/`

(4) Downstream Visualization and Biological Interpretation

After identifying SV genes, reproduce downstream analyses and visualization, including:

- Venn and UpSet plots:
 - `RealData/code/venn_plot_draw_Figures_S1-S3.R`
 - `RealData/code/upset_draw_Figures_S43-S45.R`
- Cell-type composition analysis:
 - `RealData/code/celltype_plot_draw_Figures_S30-S32.R`
- comparative analysis:
 - `RealData/code/comparative_analysis_Figures_2_S46_S47.R`

(5) Comparative Benchmarking

Next, reproduce benchmarking analyses, including:

- Comparative performance in identifying marker genes:
 - `RealData/code/marker_gene_identify_Tables_S18-S20.R`

(6) Gene Clustering and Spot Clustering

Finally, reproduce clustering analyses based on identified SV genes:

- Gene cluster inference and visualization:
 - `RealData/code/gene_cluster_draw_Figures_S48-S53_Table_S24-S26.R`
- Spot clustering and spatial domain identification:
 - `RealData/code/comparison_spotcluster_index_S27-S44.R`
 - `RealData/code/spatial_domain_draw_Figure3.R`

Key Note: All R scripts in `code/` use the `here` package to reference files in `result_data/`. Ensure the working directory is set to the `code_and_data` root before execution.